

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: Why do some numbers from our everyday Kenyan lives "end neatly" while others go on forever — and how does knowing the difference help us make better decisions?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Number Classification — What Kind of Number Is That?	Teacher presents the anchoring phenomenon: Sister Wanjiku, a community health worker in Kisumu, records patient data — temperatures (37.5 °C), medicine doses (0.75 ml), and room dimensions ($\sqrt{2}$ m ²) — and notices some numbers 'end neatly' while others do not. Students observe a display of Kenyan real-life numbers (e.g., price of ugali KES 50, maize yield 3.75 bags, circumference of a circular shamba involving π) and sort them into 'ends neatly' vs. 'goes on forever' on sticky notes posted to the class Driving Question Board (DQB). Pedagogical note: Accept all student-generated sorts without correction at this stage; the goal is to surface prior knowledge and misconceptions that will anchor subsequent lessons.	Students learned that numbers from everyday Kenyan life can be grouped based on whether their decimal representations terminate or continue indefinitely. They identified that quantities like KES 37.50 or 0.75 ml 'end neatly,' while measurements involving roots or π do not. They were introduced to the vocabulary: rational number, irrational number, and real number, understanding these as preliminary labels to be formalised in later lessons. Pedagogical note: Emphasise that the driving question is not yet answered — students should leave with productive curiosity, not closure.	Teacher explains that ALL numbers students sorted belong to the broader set called Real Numbers ($\mathbb{R}$). The 'ends neatly' group corresponds to rational numbers — expressible as p/q where p and q are integers and $q \neq 0$ — while the 'goes on forever (non-repeating)' group corresponds to irrational numbers. Sister Wanjiku's data is used as the reference: her temperature readings and doses are rational; the diagonal of a square medicine cabinet ($\sqrt{2}$) is irrational. Teacher draws the nested set diagram ($\mathbb{R} \supset \text{Rationals} \supset \text{Integers} \supset \text{Whole Numbers}$; $\mathbb{R} \supset \text{Irrationals}$) on the board and asks students to update their DQB sticky notes using correct labels. Pedagogical note: Use cold-calling (minimum 4 students) and provide 10 seconds of wait time before accepting answers to encourage deeper thinking.
Lesson 2 Observe: Rational vs. Irrational Numbers — Sorting Kenyan Real-Life Data	Teacher distributes a data card set featuring authentic Kenyan quantities: the price per kg of sukuma wiki (KES 12.50), the length of a running track in Eldoret (400 m), the diagonal of a square plot of land ($\sqrt{2}$	Students learned that rational numbers produce decimal expansions that either terminate (e.g., $12.50 = 25/2$) or repeat in a predictable pattern (e.g., $0.333... = 1/3$), while irrational numbers such as $\sqrt{2}$ ($\approx$	Teacher consolidates by revisiting Sister Wanjiku's anchoring phenomenon: her medicine doses ($0.75 \text{ ml} = 3/4$) and temperatures ($37.5 \text{ °C} = 75/2$) are rational because they satisfy the p/q definition. The

	km), the ratio of a circular water tank's circumference to diameter (π), and a repeating decimal from a bean-yield calculation (0.333...). Students work in pairs to sort cards into rational and irrational categories on a T-chart and record decimal expansions using calculators. Pedagogical note: Circulate and prompt pairs with: 'Can you write that number as a fraction of two integers?' rather than giving answers directly.	1.41421356...) and π ($\approx 3.14159265...$) produce non-terminating, non-repeating decimals. They also learned that the test 'Can it be written as p/q?' is the key criterion for classifying any real number. Pedagogical note: Highlight the bean-yield repeating decimal explicitly — many students incorrectly classify repeating decimals as irrational because they 'go on forever.'	circular shamba measurement involves π and is therefore irrational. Teacher formalises: Rational = $\{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$; Irrational = real numbers that cannot be expressed this way. Students annotate their nested set diagram from Lesson 1 with two new examples each. Pedagogical note: Ask at least 3 students to justify their classification aloud using the formal definition — correct informal language ('it has a pattern') into precise mathematical language ('it is a repeating decimal, therefore rational').
Lesson 3 Explain: Defining and Formalising Rational vs. Irrational Numbers	Teacher projects a set of numbers drawn from Kenyan contexts on the board: 7, -3, 0, $2/5$, $\sqrt{9}$, $\sqrt{5}$, 0.6 (0.666...), π, $-\sqrt{16}$, and the cost-per-person of a githeri meal split among 3 friends ($\text{KES } 50 \div 3$). Students individually write whether each is rational or irrational and give a one-sentence justification before any instruction. Pedagogical note: Collect written predictions to identify the most common misconceptions (e.g., students who classify $\sqrt{9}$ as irrational because it contains a root symbol) — these become the focus of direct instruction.	Students learned and formalised: (1) A RATIONAL number can always be written as p/q ($p, q \in \mathbb{Z}, q \neq 0$) — its decimal either terminates or repeats. (2) An IRRATIONAL number cannot be written as p/q — its decimal is non-terminating and non-repeating. (3) Perfect-square roots (e.g., $\sqrt{9} = 3$) are rational; non-perfect-square roots (e.g., $\sqrt{5}$) are irrational. (4) Every integer is a rational number (e.g., $7 = 7/1$). Pedagogical note: The githeri-splitting example ($50/3 = 16.666...$) is particularly powerful — it shows a repeating decimal arising from a real Kenyan transaction, confirming it is still rational.	Teacher works through each number from the observation set explicitly, writing the p/q form where applicable. For irrational numbers, teacher demonstrates (informally) that no fraction can exactly equal $\sqrt{5}$ by showing that squaring nearby fractions (e.g., $9/4 = 2.25$, $7/3 \approx 2.33$) never gives exactly 5. Students return to the DQB and revise any earlier sticky notes that were mis-classified. Teacher poses the bridging question: 'Now that we can classify numbers, what happens when we add, subtract, multiply, or divide them — do the results stay rational?' This sets up Lessons 4 and 5. Pedagogical note: Use wait time of at least 8 seconds after the bridging question before cold-calling 3–4 students; record their hypotheses on the DQB for revisiting in Lesson 7.
Lesson 4 Combined Operations on Rational Numbers I — Addition & Subtraction	Teacher presents a market scenario set in Kericho: a tea farmer buys fertiliser for KES $3/4$ of her weekly budget, seeds for KES $1/6$ of her budget, and wants to know the total fraction spent and how much remains. Students first estimate answers mentally, then work through fraction addition and subtraction problems drawn from Kenyan farm and household budgets (e.g., $2/3$ kg of maize flour + $1/4$ kg of wheat flour; a sukuma wiki recipe calling for $1\frac{1}{2}$ bunches minus $3/4$ used). Pedagogical note: Require students to write a sentence connecting each calculation to the real Kenyan context	Students learned that adding or subtracting rational numbers always produces a rational number (closure property). They mastered the procedure: find the Lowest Common Denominator (LCD), convert fractions, perform the operation, simplify. They also learned to add and subtract mixed numbers by converting to improper fractions first. Key insight: the result of any addition or subtraction of p/q numbers can itself be written as p/q, so the rational number set is closed under addition and subtraction. Pedagogical note: Surface the LCD method explicitly using the Kericho	Teacher formalises closure under addition and subtraction: if $a = p/q$ and $b = r/s$ (both rational), then $a + b = (ps + rq)/(qs)$, which is also p/q form, hence rational. Teacher connects back to Sister Wanjiku: when she totals two medicine doses ($0.75 \text{ ml} + 0.25 \text{ ml} = 1 \text{ ml}$), the result is rational. Students add a 'Operations on Rationals' section to their cumulative model/diagram and record: Addition ✓ Rational, Subtraction ✓ Rational. Pedagogical note: Cold-call 3 students to state the closure property in their own words before moving to consolidation exercises.

	— this combats purely procedural work and keeps the anchoring phenomenon central.	farmer's budget numbers — $\frac{3}{4} + \frac{1}{6}$ requires LCD = 12, giving $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$. Confirm the answer is rational.	
Lesson 5 Combined Operations on Rational Numbers II — Multiplication & Division	Teacher presents a nyama choma vendor in Nairobi who buys $2\frac{1}{3}$ kg of beef and needs to divide it equally among 5 plates, then multiply the cost per kg (KES 650) to find the total cost. Students also explore: if a maize bag holds $\frac{3}{4}$ of a tonne and a farmer has $2\frac{1}{2}$ bags, how many tonnes does she have? Students predict outcomes and attempt calculations before formal instruction. Pedagogical note: Deliberately include a division-by-fraction problem (e.g., how many $\frac{3}{4}$ -kg portions fit in $4\frac{1}{2}$ kg of ugali dough?) to surface the common 'invert and multiply' misconception — many students apply it without understanding why.	Students learned that multiplying rational numbers involves multiplying numerators together and denominators together: $(\frac{p}{q}) \times (\frac{r}{s}) = \frac{pr}{qs}$, which is rational. Dividing rational numbers means multiplying by the reciprocal: $(\frac{p}{q}) \div (\frac{r}{s}) = (\frac{p}{q}) \times (\frac{s}{r}) = \frac{ps}{qr}$, also rational. Mixed numbers must be converted to improper fractions before operating. Key insight: the rational number set is also closed under multiplication and division (excluding division by zero). Pedagogical note: Stress WHY the 'invert and multiply' rule works — dividing by a fraction is the same as multiplying by its reciprocal because $\frac{r}{s} \times \frac{s}{r} = 1$.	Teacher consolidates the four operations together: all four (addition, subtraction, multiplication, division) preserve rationality. Students update their cumulative model to include: Multiplication ✓ Rational, Division ✓ Rational ($\div 0$ undefined). Teacher connects back to the anchoring phenomenon: Sister Wanjiku multiplies a dose ($\frac{3}{4}$ ml) by a frequency (4 times/day) to get 3 ml/day — rational. The nyama choma vendor's per-plate cost is rational. Teacher poses: 'What if we tried to operate on irrational numbers — would the result always be irrational?' This is recorded as a new DQB question. Pedagogical note: Allow 2 minutes of pair discussion before cold-calling 3 students for predictions.
Lesson 6 Number Patterns, Sequences & Rational Approximations of Irrationals	Teacher shows students a sequence of rational approximations of π used historically by mathematicians ($\frac{22}{7}$, $\frac{333}{106}$, $\frac{355}{113}$) and asks: 'Which is closest to π , and how would a Kenyan engineer designing a circular water tank in Mombasa decide which approximation is good enough?' Students also examine a sequence of rational numbers converging to $\sqrt{2}$ (1, 1.4, 1.41, 1.414, 1.4142...) using a calculator, recording each term and plotting on a number line. Pedagogical note: Emphasise that although $\sqrt{2}$ and π are irrational, rational numbers can get arbitrarily close — this motivates the concept of density of rationals and practical approximation in engineering contexts.	Students learned that irrational numbers can be approximated to any desired degree of accuracy using rational numbers, and that choosing the right level of precision is a practical decision (e.g., a carpenter in the Rift Valley cutting timber needs $\sqrt{2} \approx 1.414$, not 100 decimal places). They also reviewed arithmetic and geometric sequences with rational terms, identifying common differences and common ratios, and confirming that rational sequences produce rational terms. Pedagogical note: Link the $\frac{22}{7}$ approximation to students' future work in mensuration — make explicit that $\frac{22}{7}$ is rational ($\frac{p}{q}$ form) but π is not, so $\frac{22}{7}$ is always an approximation.	Teacher formalises: on the real number line, between any two rational numbers there is always another rational number (density property), and irrational numbers fill the 'gaps' that rationals cannot reach exactly. Teacher demonstrates with the number line: 1.414 and 1.415 are both rational; $\sqrt{2}$ lies between them but equals neither. Students annotate their cumulative number-line model with rational approximations of $\sqrt{2}$ and π . Teacher updates the DQB with student-generated answers about why approximations are 'good enough' for Kenyan practical contexts. Pedagogical note: Cold-call students to name one real Kenyan scenario where a rational approximation of an irrational number is used — accept and discuss at least 4 examples.
Lesson 7 Real Numbers in Real-Life Applications & DQB Consolidation	Teacher presents a multi-context Kenyan problem set: (a) Sister Wanjiku records patient temperatures over a week and calculates the mean; (b) a githeri seller in Kisumu calculates profit margins using fractions of cost price; (c) a farmer in the	Students confirmed that numbers arising from everyday Kenyan contexts — health temperatures, food costs, farm masses — are rational because they can be expressed as $\frac{p}{q}$ and their decimal expansions terminate or repeat. They also confirmed	Teacher leads a whole-class DQB review: reads each original question posted, invites student volunteers to provide a now-justified answer, and marks questions as 'Answered ✓', 'Partially Answered ⚡', or 'Still Investigating ?'. Teacher returns to the

	Rift Valley computes the area of a circular borehole cross-section using π. Students work in groups to classify all numbers arising in each scenario, perform combined operations, and state whether each result is rational or irrational. Pedagogical note: Assign each group a different scenario, then conduct a gallery walk so all students engage with all three contexts before whole-class consolidation.	that geometric calculations involving circles or diagonals of squares introduce irrational numbers, but rational approximations are used in practice. Students synthesised all four operations and their closure properties, and revisited every DQB question posted since Lesson 1, providing evidence-based answers. Pedagogical note: Explicitly celebrate growth in understanding by projecting the DQB from Lesson 1 and Lesson 7 side-by-side — students should see how their initial predictions have been refined by evidence.	anchoring phenomenon: Sister Wanjiku's data is revisited in full — every number is classified, every operation is verified. Teacher states: 'We can now answer our driving question — numbers from everyday Kenyan life end neatly because they arise from counting and measuring with fractions; irrational numbers arise from geometry and certain roots, and knowing the difference helps us choose the right level of precision for real decisions.' Students write a one-paragraph personal answer to the driving question in their exercise books. Pedagogical note: Collect and review paragraphs as formative assessment before Lesson 8.
Lesson 8 Summative Model Revision, Assessment & Sub-Strand Closure	Teacher returns students' Lesson 7 driving-question paragraphs with brief written feedback. Students are given a final model-revision task: update their cumulative nested set diagram and number-line model to include all vocabulary, examples, and operation closure properties learned across the sub-strand. They then complete a structured summative task using a new Kenyan scenario: a Nairobi posho mill owner records batch masses (rational), calculates circular sieve areas (irrational approximated), splits profits as fractions (rational operations), and sequences production over weeks (rational pattern). Students classify, operate, and justify in writing. Pedagogical note: The summative task is designed to require ALL skills from Lessons 1–7 in a single coherent context — resist breaking it into isolated sub-questions; require students to move fluidly between classification, operations, and justification.	Students demonstrated mastery of the full sub-strand: defining and classifying real, rational, and irrational numbers; performing all four combined operations on rational numbers; approximating irrationals with rationals; identifying rational sequences; and applying these skills to authentic Kenyan contexts. They revised their models to reflect complete, accurate understanding — replacing early misconceptions (e.g., 'repeating decimals are irrational') with correct, evidence-supported knowledge. Pedagogical note: Use the final models as portfolio evidence of conceptual growth; compare each student's Lesson 1 sticky-note sorts with their Lesson 8 annotated diagram as a visible record of learning.	Teacher facilitates a sub-strand closure discussion: 'Why does it matter — in a Kenyan hospital, market, or farm — whether a number is rational or irrational?' Students share responses; teacher synthesises by affirming that rational numbers allow exact calculation and fair division (critical for medicine doses, market prices, and land measurement), while recognising irrational numbers prevents false precision (a farmer should not claim to know $\sqrt{2}$ exactly when measuring a diagonal fence). Teacher previews the next sub-strand, connecting real numbers to algebraic expressions. Pedagogical note: End with a class 'Number of the Day' celebration — post a Kenyan real-life number on the board and have the class, in unison, classify it as rational or irrational with a thumbs up/down signal, consolidating the sub-strand's core competency in a memorable, low-stakes way.